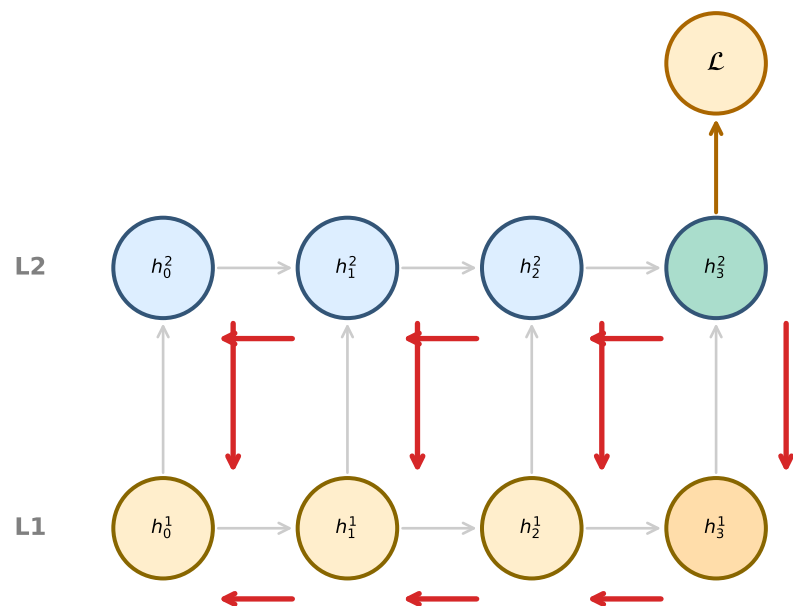


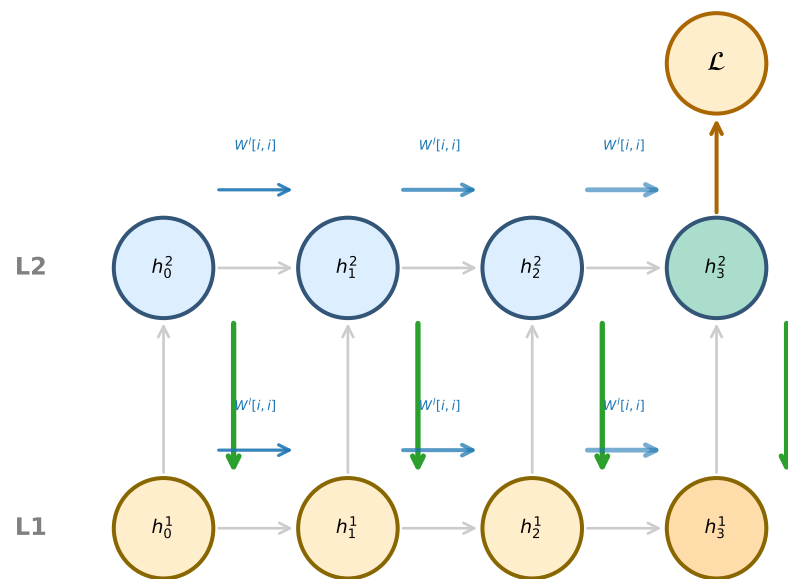
Credit assignment in a 2-layer deep RNN: BPTT vs deep e-prop vs d=0

BPTT



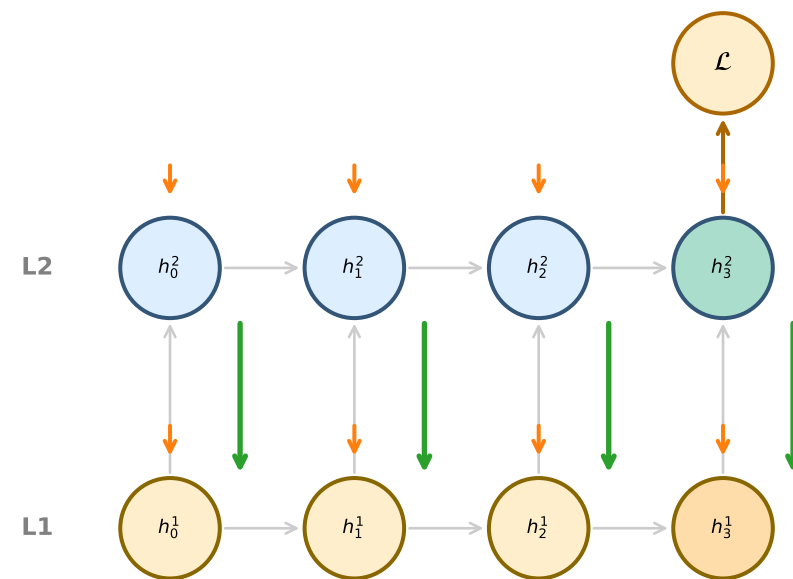
Temporal: exact backward at each layer (red $\leftarrow$)
 Spatial: exact backward through layers (red $\downarrow$)

Deep E-prop



Temporal: diagonal carry $W'[i, i]$ (blue $\rightarrow$, approximate)
 Spatial: exact Jacobian J_{ff} (green $\downarrow$)

d=0



Temporal: no carry — instantaneous only (orange $\downarrow$)
 Spatial: exact Jacobian J_{ff} (green $\downarrow$)